What are Outliers

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Where this fits: Pipeline map step 4 (Clean). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	**4 Clean**	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Poor-quality data; Univariate analysis.
- **Used here, taught in full later:** Percentile outlier method.
- **Leads to:** Z-score outlier method; IQR outlier method; Percentile outlier method.
- **Compare with:** Missing values.

1. Overview

Key point: An **outlier** (G-1420) is a value far away from the rest of the data; handling outliers means three steps: decide whether to keep them, detect them, then treat them.

Outliers came up briefly in the box plot's whiskers and outliers (the dots beyond a box plot's fences) and in detecting and removing outliers (a few points pulling a regression line). This Note opens a group of four Notes on them. Here we learn what outliers are, why they cause trouble, when to keep them, and which techniques the next Notes use.

Figure 1 is the map of the whole group. First we decide whether the outliers are errors or real, useful values. If we treat them, we first **detect** them by setting a lower and an upper limit, then **treat** them by removing or changing the values outside the limits.

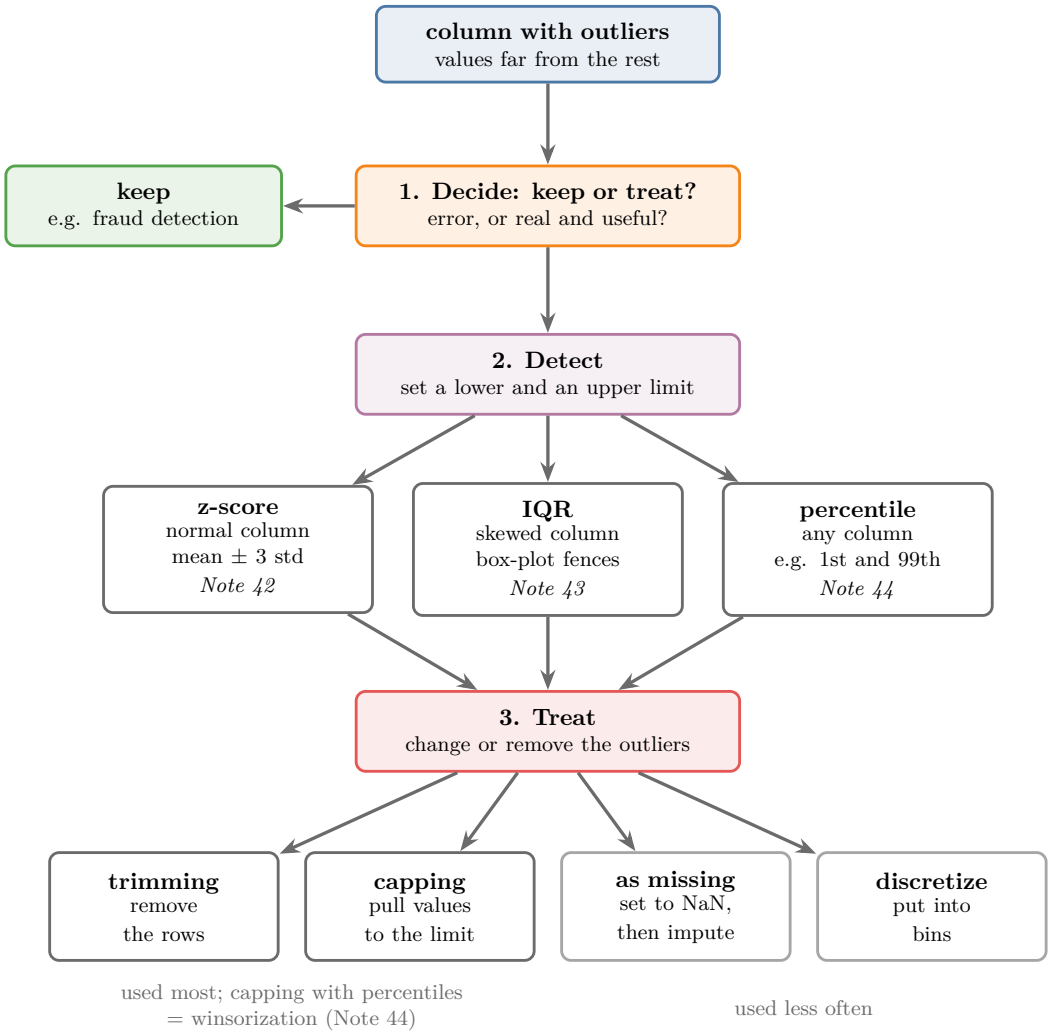


Figure 1: Handling outliers: decide, detect with one of three rules, then treat

2. What an outlier is

> **Key point:** An outlier is a data point that is very different from all the other data points.

Every class has one student who scores top marks while the rest of the class struggles to pass. That student is an outlier of the class: nothing like everyone else.

In data, an **outlier** (G-1420) is a data point (an **observation** (G-1374): one record, one row of the data table) that lies very far from the other data points. Each **feature** (G-772) is an input variable (one column of the table), and the **target** (G-1949) is the output a model predicts. An outlier can be very high or very low, but never in the middle: an average value is by definition not an outlier.

2.1 One outlier changes the mean

> **Key point:** A single extreme value can move the mean so far that it no longer describes anyone.

Take a class of nine people with salaries between 15,000 and 22,000 rupees a month. Now a billionaire, such as Bill Gates, joins the class. The class's mean salary jumps into the crores, a number that describes nobody in the room.

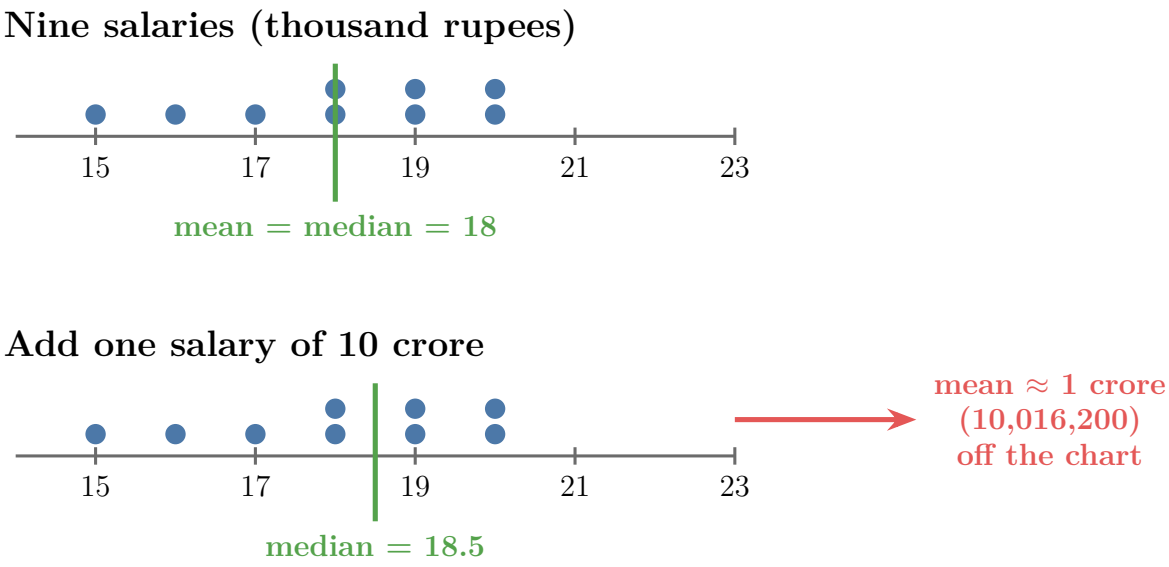


Figure 2: Nine salaries, then the same nine plus one of 10 crore: the mean leaves the chart, the median moves by 500 rupees

Figure 2 draws the numbers worked out in the box below: watch the mean fly off while the median stays among the nine.

> **Extra:** The numbers, step by step.
>
> 1. **In words:** the mean is the sum of the values divided by how many there are. One huge value makes the sum huge.
> 2. **Formula:**
> $$\text{mean} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$
> 3. **Example:** take the nine salaries 15,000, 16,000, 17,000, 18,000, 18,000, 19,000, 19,000, 20,000 and 20,000 rupees. They add up to 162,000 rupees (mean 18,000). Add one salary of 100,000,000 rupees (10 crore):
> $$\text{mean} = \frac{162{,}000 + 100{,}000{,}000}{10}$$
> $$\text{mean} = 10{,}016{,}200$$

The mean is now about 1 crore. The median of the ten salaries is the mean of the two middle values:

$$(18{,}000 + 19{,}000)/2 = 18{,}500$$

That is 18,500 rupees, against 18,000 for the nine, so the median barely moves.

One extreme value can quietly spoil a whole analysis in this way. In ML, **outliers** (G-1420) are therefore handled with care: removed, or changed, or kept on purpose.

3. How outliers spoil a model

Key point: Linear regression draws its line close to every point, so a few **outliers** (G-1420) pull the line away from the pattern of all the others.

A regression line pulled by outliers is drawn in detecting and removing outliers. Here we keep one example in mind: students' weekly study hours against their marks, where more hours bring more marks.

Two students studied very little and still scored top marks. These two outliers tilt a **linear regression** (G-1094) line towards them, so it fits the other students worse. Outliers are dangerous because they are often hidden: nothing warns us, the model just performs worse.

Figure 3 puts numbers on it with example data for 20 students. The line through them rises 3.33 marks per study hour and misses a typical student by 4.5 marks. Adding the two outliers flattens the slope to 1.91, and the line now misses the same 20 students by 8.4 marks on average, almost twice as much. The line is chosen to keep the total squared distance to all points small, so two far points pull it towards themselves.

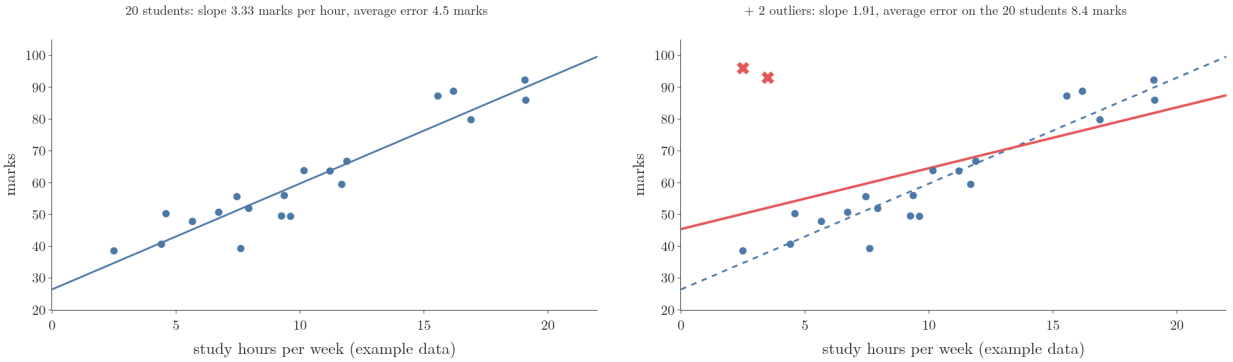


Figure 3: Example data: 20 students' study hours and marks, with the least-squares line (blue). Adding two students with few hours and top marks (red crosses) tilts the line (red) away from the other 20.

4. When to remove and when to keep outliers

> **Key point:** Remove an **outlier** (G-1420) when it is an error; keep it when it is real and is exactly what the problem is about.

Outliers are not always harmful. Deciding whether an outlier is dangerous or a genuine part of the data is an important skill.

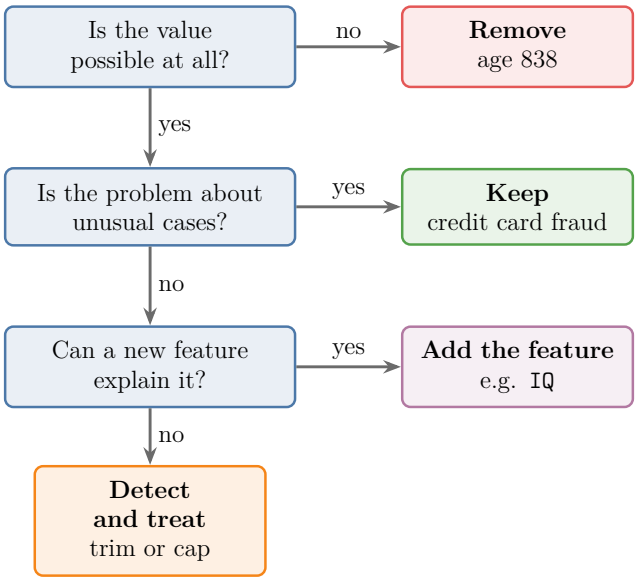


Figure 4: The three cases of this section as one decision path

Figure 4 puts Sections 4.1 to 4.3 in order as questions to ask about an outlier.

4.1 Remove: errors in the data

> **Key point:** A value that is impossible, such as an age of 838, is an error and should go.

Suppose we collect people's ages through an online form, and one person types 838. No human is 838 years old, so the value is wrong. The person may have meant 38 or 83, but we have no way to tell.

Since the true value cannot be recovered, the best choice is to remove it (or treat it as a missing value, Section 7.3).

4.2 Keep: when outliers are the point

> **Key point:** In anomaly detection, such as spotting credit card fraud, the **outliers** (G-1420) are exactly what we are looking for.

Some problems are about finding unusual cases. **Anomaly detection** (G-201) algorithms search for data points that do not behave like the rest. A common example is fraud detection: finding the few credit card transactions that look wrong among millions of normal ones.

There, the fraudulent transactions are the outliers. If we removed the outliers, we would remove the very thing the model must learn to find.

4.3 Explain: outliers can point to a missing feature

> **Key point:** Sometimes an **outlier** (G-1420) is real and makes sense once we add a feature that explains it.

Go back to the two students of Section 3, with few hours and top marks. They are real students, not errors. Instead of deleting them, we could add a **feature** (G-772) such as `IQ`.

With that feature, the model may learn why these two scored high with few hours, and they stop looking like outliers. An outlier can tell us that our data is missing something.

4.4 The hard part: deciding what to do

> **Key point:** Finding an **outlier** (G-1420) is easy; deciding whether to keep, remove or change it takes thought about the problem.

Spotting outliers is usually simple; the rules in Section 8 do it. The hard part is the decision, and it depends on:

- **the problem:** is it about typical cases, or about unusual ones?
- **how the outlier got into the data:** a typing error, a broken sensor, or a real but rare case?

5. Which algorithms outliers affect

> **Key point:** Algorithms that learn weights (linear and logistic regression, AdaBoost, deep learning) are strongly affected; tree-based algorithms hardly are.

Not every algorithm reacts to **outliers** (G-1420). A simple rule of thumb: if the algorithm computes **weights** (coefficients, G-407; one number per feature, learned from all the points), outliers affect it. Squared-error loss, used by **linear regression** (G-1094), and the exponential loss of AdaBoost both give the largest errors the most say (ESL §10.6), and neural networks rate poorly on robustness to outliers in the inputs (ESL Table 10.1). A tug-of-war is the picture: every point pulls on the line, and a point far away pulls hardest.

A small example with numbers:

1. The points (1, 1), (2, 2) and (3, 3) lie on a line with slope 1.
2. Add one far point, (4, 20).
3. The line that stays closest to all four points in the squared-error sense now has slope 5.8, nearly six times steeper, although three of the four points did not move.

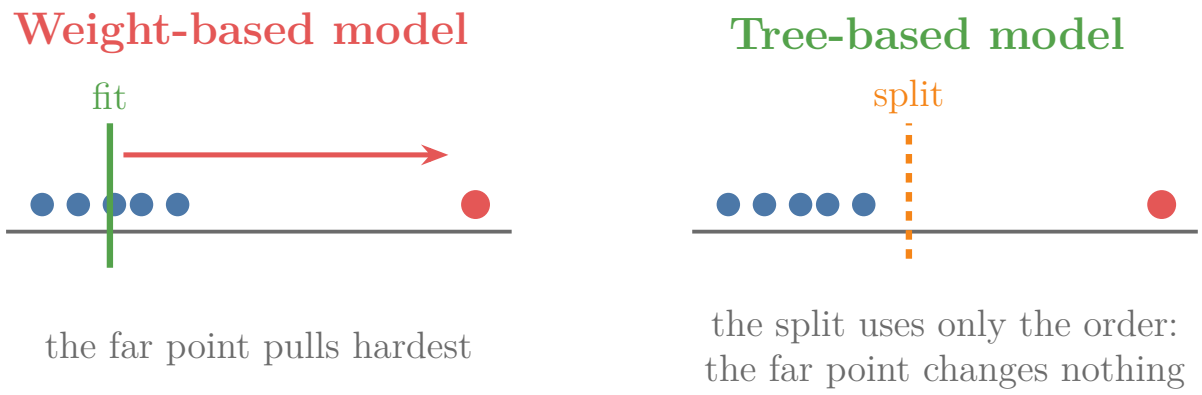


Figure 5: A weight-based fit is pulled by a far point; a tree's split depends only on the order of the values

Figure 5 shows the two behaviours side by side.

Affected strongly	Hardly affected
linear regression	decision tree
logistic regression	random forest
AdaBoost	gradient boosting
deep learning (neural networks)	other tree-based algorithms

Tree-based algorithms cut the input space into regions with simple conditions, such as "hours < 5". Such a condition depends only on the order of the values, not on how far out an extreme value lies, so outliers in the inputs barely change it (ESL §10.7, Table 10.1).

In practice we usually try several algorithms on one problem, including weight-based ones. So treating outliers before training is a good habit either way.

Extra: A few more cases.

- **Also affected:** k-means, because squaring the distances gives the largest distances the most say (ESL §14.3.10); SVMs (ESL Table 10.1); and PCA, which is built from the mean and covariance of the data (Hubert et al. 2005). Scaling with the mean and standard deviation (standardization, see the standardization formula) is pulled too, since one extreme value moves the mean (Section 2.1; scikit-learn docs, Compare the effect of different scalers).
- **Trees are not fully immune:** an outlier in the ***target*** *(G-1949)* still shifts the average that a regression tree predicts in its leaf. Gradient boosting with the usual squared-error loss is pulled by such outputs; using the absolute error or the Huber loss instead makes it robust (ESL §10.6, §10.9). In scikit-learn this is `GradientBoostingRegressor(loss="huber")`.

6. Handling outliers: detect, then treat

> **Key point:** We first set limits that separate normal values from **outliers** (G-1420), then deal with the values outside the limits.

Once we decide to handle outliers, the work has two parts (Figure 1):

1. **Detection:** compute a lower and an upper limit for the **feature** (G-772). Values outside them are outliers.
2. **Treatment:** remove or change those values.

Section 7 maps the treatments, Section 8 the detection rules. Notes ML-041 to ML-043 then put each rule to work (**z-score** (G-2141), IQR, percentiles).

Figure 6 runs both parts on the 714 known Titanic ages, with the IQR fences of section 8.2 as the detection rule. The **IQR** (interquartile range, G-966) is the width of the middle half of the data: the 75th percentile Q3 minus the 25th percentile Q1. Each fence lies 1.5 IQR beyond Q1 or Q3:

$$Q1 = 20.125$$

$$Q3 = 38$$

$$\text{IQR} = 38 - 20.125 = 17.875$$

$$1.5 \times 17.875 = 26.81$$

$$\text{lower} = 20.125 - 26.81 = -6.69$$

$$\text{upper} = 38 + 26.81 = 64.81$$

No age is below the lower fence, and 11 ages are above the upper one. **Capping** (G-345) (section 7.2) then moves those 11 ages onto the fence and keeps every row.

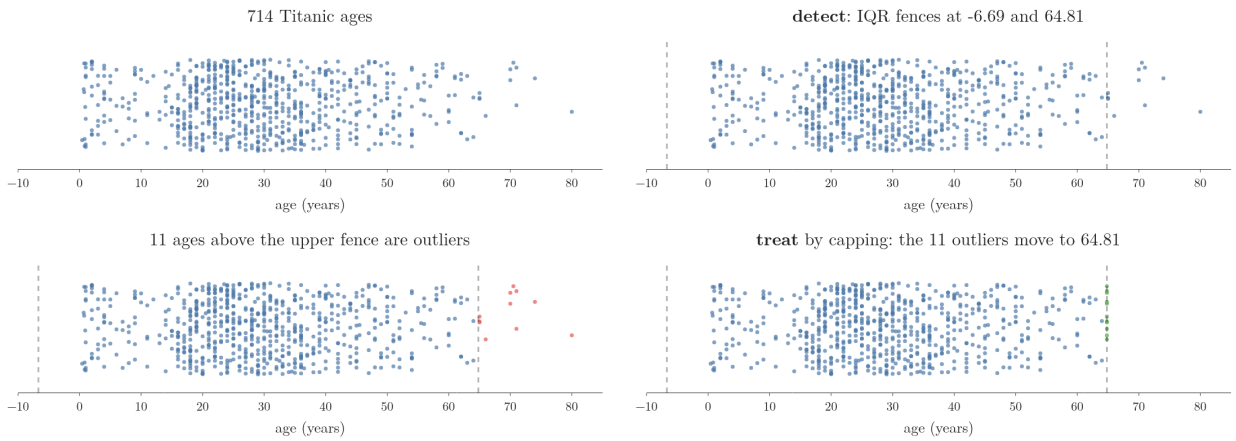


Figure 6: The 714 known Titanic ages: detect (IQR fences, dashed), mark the 11 outliers above 64.81 (red), then treat them by capping onto the fence (green).

7. Ways to treat outliers

> **Key point:** Trimming removes the **outliers** (G-1420), capping pulls them back to the limit; treating them as missing values and discretizing them are used less often.

7.1 Trimming

> **Key point:** Trimming deletes the **observations** (G-1374) that hold outliers; it is fast, but too much trimming shrinks the data.

Trimming (G-2019) removes every observation whose value lies outside the limits. In Section 3 we would simply delete the two students.

- **Advantage:** very fast and simple.
- **Disadvantage:** if there are many **outliers** (G-1420), we delete many observations, and the data gets thin.

7.2 Capping

> **Key point:** Capping keeps the **observations** (G-1374) but replaces every value beyond a limit with the limit itself.

Outliers (G-1420) always sit at an end of the data, either too high or too low. **Capping** (G-345) sets the two limits and moves every value beyond them back onto the limit. Figure 7 compares it with **trimming** (G-2019), with limits 20 and 90.

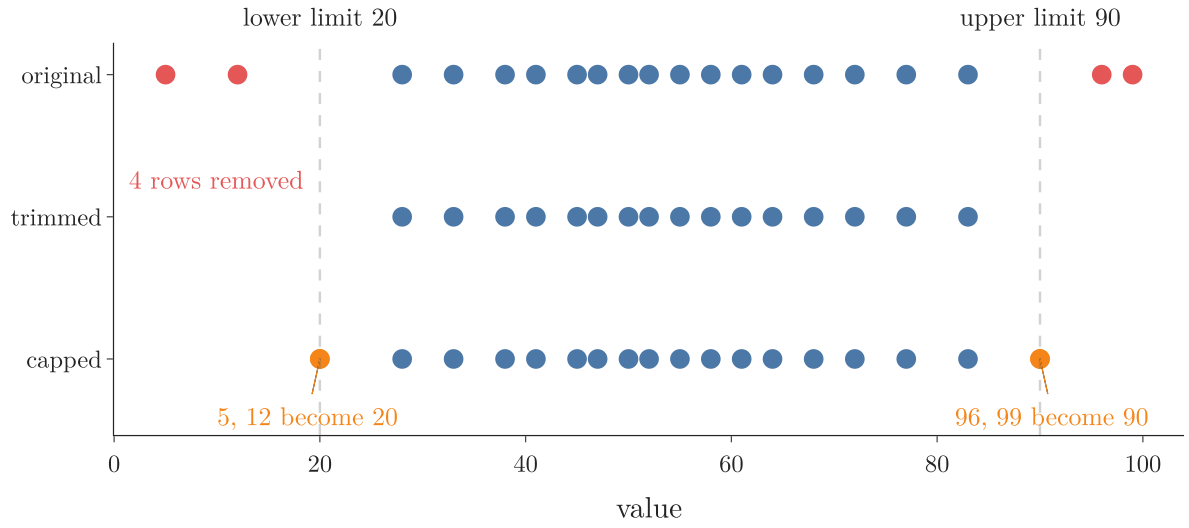


Figure 7: Trimming deletes the four outliers; capping keeps all 20 rows and moves the outliers onto the limits

- Values below 20 (here 5 and 12) become 20.
- Values above 90 (here 96 and 99) become 90.
- All 20 observations stay; with trimming only 16 remain.

Capping with limits set by percentiles is called **winsorization** (G-2123); The percentile rule covers it.

7.3 Two less common treatments

> **Key point:** An **outlier** (G-1420) can also be treated as a missing value, or hidden inside a range by discretization.

- **Treat as missing:** replace each outlier with `NaN` and then fill it in with any imputation technique, from mean and median imputation to iterative imputation.
- **Discretization** (G-619), or binning: turn the numbers into ranges, such as 0 to 10, 10 to 20, . . . , 90 to 100 (see **discretization** (G-619)). An extreme value of 99 just falls into the last range with all the other high values, so it no longer stands out.

Trimming (G-2019) and capping are used far more, and they are what Notes ML-041 to ML-043 focus on.

8. Ways to detect outliers

> **Key point:** The **feature** (G-772)'s shape decides the rule: mean ± 3 standard deviations for a normal feature, the IQR fences for a skewed one, percentiles for any feature.

Many detection methods exist; these three are the most important. Figure 8 applies each to example data, with the limits as dashed lines and the flagged values in red.

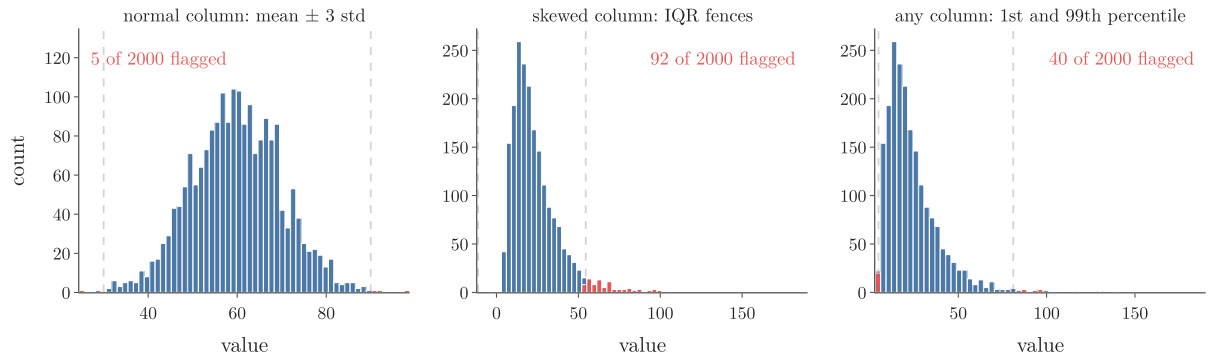


Figure 8: The three detection rules: mean plus or minus 3 standard deviations, the IQR fences, and the 1st and 99th percentiles

8.1 Normal feature: mean ± 3 standard deviations

> **Key point:** In a normal feature, a value more than 3 standard deviations from the mean is an **outlier** (G-1420).

About 99.7% of a normal **feature** (G-772)'s values lie within 3 standard deviations of the mean, so a value outside that range is rare enough to call an outlier. The 68-95-99.7 rule explains the rule and its limits.

8.2 Skewed feature: the IQR fences

> **Key point:** For a skewed **feature** (G-772), the box-plot fences, 1.5 IQR beyond the box, set the limits.

The box-plot fences, 1.5 IQR (the width of the middle half of the data) below Q1 and above Q3, as computed for the Titanic ages in Section 6 (whiskers and outliers) serve as the limits for a skewed feature; the fences puts them to work.

8.3 Any feature: percentiles

> **Key point:** The percentile rule calls everything below a low percentile or above a high percentile an **outlier** (G-1420); it works for any shape.

The third rule works whatever the shape of the **feature** (G-772). We pick two percentiles, for example the 1st and the 99th, and every value below the first or above the second is an outlier.

The cut-offs are our choice, depending on the problem: 1 and 99, 2.5 and 97.5, or 5 and 95. With 1 and 99, about 2% of the values are always flagged (40 of 2,000 in Figure 8, right). The percentile rule covers this rule together with **winsorization** (G-2123).

9. The outlier Notes, in order

> **Key point:** Notes ML-041 to ML-043 each take one detection rule and use it to trim and cap.

Note	Detection rule	Use when
z-score (ML-041) (G-2141)	z-score: mean ± 3 standard deviations	the column is roughly normal
IQR (ML-042)	IQR: box-plot fences	the column is skewed
percentiles (ML-043)	percentiles, and winsorization	any column

10. Summary

Step	Options	Notes
Decide	remove errors; keep outliers the problem is about; add a feature that explains them	this Note
Detect	mean ± 3 std (normal), IQR fences (skewed), percentiles (any)	ML-041, ML-042, ML-043
Treat	trimming, capping (winsorization); less often: as missing, discretization	ML-041 to ML-043

- An **outlier** (G-1420) is a data point very different from the rest; it lies at the high or low end, never in the middle, which is why the detection rules set a lower and an upper limit.
- One outlier can move the mean far from every real value (one 10-crore salary lifts the class mean to about 1 crore); the median barely moves, because it only looks at the middle values.
- A few outliers can pull a **linear regression** (G-1094) line away from the pattern of all other points, because the line keeps the total squared distance small and far points pull hardest: two outliers almost doubled the line's typical miss on 20 students.
- Remove outliers that are errors, because their true value cannot be recovered; keep them when they are the point, as in fraud detection, because removing them removes what the model must find.
- Weight-based algorithms (linear and logistic regression, AdaBoost, deep learning) are sensitive to outliers, because their losses give the largest errors the most say; tree-based ones hardly are, because a split such as "hours < 5" depends only on the order of the values.
- Trimming deletes outlier **observations** (G-1374) (fast, but loses data); capping moves outliers onto the limits (keeps every observation), so capping avoids the thin data that heavy trimming leaves.
- So handling outliers means deciding whether they are errors or the point of the problem, then detecting them with limits, then **trimming** (G-2019) or capping the values beyond those limits.

11. Sources

Built from

- CampusX, "What are Outliers | Outliers in Machine Learning", YouTube, https://www.youtube.com/watch?v=Lln1PKgGr_M

Other references

- ESL: Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. Springer. Section 10.7 and Table 10.1 (trees, neural networks and SVMs and outliers in the inputs), Section 10.6 (squared-error and exponential losses are not robust) and Section 10.9 (robust losses for boosting), Section 14.3.10 (k-means and outliers).
- Hubert, M., Rousseeuw, P. J. and Vanden Branden, K. (2005). ROBPCA: A New Approach to Robust Principal Component Analysis. *Technometrics* 47(1), 64–79.
- scikit-learn example, *Compare the effect of different scalers on data with outliers*.

12. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Capping (G-345)	Replacing every value beyond a limit with the limit itself.
Trimming (G-2019)	Removing the rows that hold outliers.
Fraud detection (G-804)	Spotting dishonest transactions; here the outliers are what we want to find.
Outlier detection (G-1419)	Finding the values that lie far from the rest of a feature: compute a lower and an upper limit, and flag every value outside them as an outlier, ready to be removed or changed.
Weight-based algorithm (G-2112)	An algorithm that learns one number per input column from all the points; sensitive to outliers.
Outlier (G-1420)	A value far from the rest of the data.
Observation (G-1374)	One record of a dataset: one row of the data table.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Target (G-1949)	The output column we predict, such as the class; also called the label.
Linear regression (G-1094)	A model that predicts a number from the inputs by fitting the straight line (or, with several inputs, the plane) that runs closest to all the training points.
Anomaly detection (G-201)	Finding rows that do not fit the pattern of the rest.
Coefficient (β_i) (G-407)	The weight of one input column: the change in the output per unit of that input, others fixed.

Term	Meaning
Z-score (G-2141)	A value after standardization: how many standard deviations it lies from the mean.
Interquartile range (IQR) (G-966)	The width of the middle half of the data: Q3 - Q1.
Winsorization (G-2123)	Handling outliers by capping them at percentile limits: values beyond a chosen low or high percentile are replaced by that percentile's value.
Discretization (G-619)	Turning a continuous column into a discrete one by cutting its range into intervals.
Tree-based algorithm (G-2013)	An algorithm that splits the data with simple conditions; hardly affected by outliers.
Percentile method (percentile rule) (G-1481)	A way to find outliers: flag values below a low percentile or above a high one (e.g. 1st and 99th); it works for any column.